Table A1.1 Unfired heat exchanger types – summary of features

Exchanger type		Maximum operating conditions: Temperature (°C)	Maximum operating conditions: Pressure (bar)	Typical surface area range for a single unit (m^2)	Comments[9]
Shell and tube		−200 to 700	350[1]	5–1000	Versatile and suitable for almost all applications, irrespective of duty, pressure and temperature (which is limited by metallurgical considerations).
Double pipe		−200 to 700	350[2]	0.25–200	Especially suited to small capacities, countercurrent operation and high pressures. Standard sections available.
Air cooler		700	350[2]	5–400[3]	Standardised bundles available. Suitable for almost all cooling duties above atmospheric temperature, irrespective of pressure and temperature (which is limited by metallurgical considerations).
Heat pipe		−100 to 400[5]	40[6]	100–1000	Ideally suited to heat exchange between large volumes of gas at low pressure. Extended surfaces may be employed on both gas sides.
Gasketed plate		−40 to 180[7]	30	1–1200	High thermal efficiency, flexible, low fouling tendency, compact, low weight, no inter-stream mixing or vibration, ease of maintenance. Usually cheapest exchanger type if suitable for operating conditions. Gasket material may inhibit use for certain fluids.
Gasketless plate		−200 to 980 according to type	As S & T type. max. ΔP between fluids = 40–80 bar according to type	Up to 10 000	Alternative to shell-and-tube and gasketed-plate types provided chemical cleaning is acceptable. Application may be limited by pressure difference between fluids.
Spiral		400	20	0.5–350	
Lamella	PTFE gasket	220	35 (dia. = 300 mm)	1–1000	High thermal efficiency, low fouling tendency, ease of maintenance. Handles suspensions, slurries and fibrous liquids.
	Asbestos gasket	500	10 (dia. = 1000 mm)		
Plate fin		−260 to 650	25[4]	Up to 1230 m^2 per m^3 volume	Compact, low weight, excellent for cryogenic applications. Multiple streams may be handled.

Table A1.1 (*cont.*)

Exchanger type		Maximum operating conditions: Temperature (°C)	Maximum operating conditions: Pressure (bar)	Typical surface area range for a single unit (m^2)	Comments[9]
Embossed plate		350	18[8]	Standard panels up to 1200 × 3600 mm	Compact and low weight. Curved units available. Especially suited to vessel jackets and immersion heaters
Graphite	Shell and tube	180	6	1.6–1650	Exceptional corrosion resistance, low fouling tendency, but limited with regard to temperature and pressure. Limited in capacity except for graphite shell-and-tube and multi-block types.
	Cubic and rectangular	180	5.2	0.65–153	
	Multi-block	180	6	0.22–240	
	Cartridge	180	6	0.16–18.6	
Glass	Shell and tube	175	5	6.3–26	
Teflon™	Shell and tube	20	10	9.3–50	
	Supercoils	50	6.6	2.3–10.5	
	Slimline coils	100	3.8	up to 23.7	
	Tank coils	150	2.2	56.9–148	
Rotary regenerators	Aluminium rotor	200	Near atmospheric	Rotor dia. = 6 m	Compact, but limited to heat exchange between low-pressure gases, even at high temperature. Subject to some inter-stream leakage.
	Steel rotor	425			
	Stainless or Incoloy™ rotor	980			

Notes:

(1) Typical upper limit, but higher pressures possible depending on diameters and closures required.
(2) Typical upper limit, but designs for higher pressures available.
(3) Shop-built bundle – larger sizes available if site built.
(4) Pressure reversals may reduce this value to 11 bar. Up to 80 bar available depending on cross-section.
(5) Higher temperatures possible with special working fluids.
(6) Typical only for working fluid, i.e. steam at approx. 250 °C.
(7) 260 °C for compressed asbestos fibre gaskets.
(8) 7 bar at 165 °C for titanium.
(9) For all heating duties, consider electrically heated exchangers, particularly if proposed heating medium is fouling, or suitable medium not readily available.